

Thank you for the detailed progress report. This version is much more complete than the previous one, and I can see that you are now trying to support your design choices with quantitative evidence. However, before moving to the next scenario or to Transfer Learning, there are several fundamental issues that need to be confirmed carefully.

Please check the following points **step by step and in this order**. Do not move forward by assuming that the previous step is correct.

Step 1. First, verify the original Nyquist data before discussing interpolation

My first concern is Figures 3 and 4.

The Nyquist plots look unusually irregular, with many sharp turns, crossings, and oscillations. Before comparing Linear, Cubic Spline, and PCHIP, please first confirm whether the **original EIS points themselves are being plotted correctly**.

Please answer clearly:

1. What is the ground-truth or reference Nyquist data in Figures 3 and 4?
2. Which points or curves come directly from the NASA measurements?
3. Which curves are generated by interpolation?
4. In what order are the raw EIS points connected?

At present, Figure 4 appears to use the black markers as the raw measured EIS points, while Linear, Cubic, and PCHIP are interpolation results. Therefore, none of the interpolated curves should be treated as the physical ground truth.

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Please make this distinction very clear.

Step 2. Confirm where the actual frequency information comes from

This is currently one of the most important questions.

Your report compares:

- Linear frequency grid
- Logarithmic frequency grid

and then selects Log-PCHIP.

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However, please confirm exactly where the **actual EIS measurement frequency vector** comes from.

The proper EIS information should conceptually be:

$$\{f_i, \operatorname{Re}(Z_i), \operatorname{Im}(Z_i)\}$$

Therefore, please go back to the original NASA `.mat` files and identify the actual frequency field used for each impedance measurement.

In the next report, please show explicitly:

Raw Frequency → Re(Z), Im(Z) → Frequency Ordering → Interpolation → 256 Features

If the current “log-frequency grid” is only generated from the index of the EIS points rather than the actual measured frequencies, then it should not be called true log-frequency interpolation, and we should not claim that it preserves low-frequency information.

Step 3. Confirm that the 39 raw EIS points are ordered by actual measurement frequency

This must be verified before any interpolation analysis.

Please check whether the 39 EIS points are connected according to:

| actual measurement frequency order

or simply:

| CSV row/index order.

If the points are connected in the wrong order, the Nyquist plot may naturally become highly irregular, with loops and crossings.

Therefore, before concluding that Cubic Spline creates an incorrect curve, first demonstrate that the **raw measured EIS points are correctly ordered by frequency**.

Please show one example battery/cycle in a table containing:

Point	Frequency	Re(Z)	-Im(Z)
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and then plot the Nyquist curve using exactly that frequency ordering.

This should be completed before proceeding with the interpolation conclusion.

Step 4. Clarify what the “standard answer” is for the Nyquist curve

There is currently no independent continuous “standard Nyquist curve” shown in Figures 3 and 4.

The measured EIS points should be treated as the experimental reference.

The Linear, Cubic Spline, and PCHIP curves are only different mathematical methods for connecting or resampling those measured points.

Therefore, please do not use PCHIP itself as the ground truth.

If you want to define a continuous physical reference curve, you would need an independent basis, for example:

- a much denser measured frequency sweep, or
- a validated equivalent-circuit-model fitting result.

Otherwise, the correct wording is simply:

PCHIP produces fewer interpolation artifacts relative to the measured EIS points.

Step 5. Recheck Figure 6 and clearly define its reference

Figure 6 compares the raw/interpolated EIS with the AE reconstructed EIS. The report concludes that the unseen B0018 reconstruction is strongly distorted.

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However, please clearly state that the black curve in Figure 6 is already the **preprocessed/interpolated AE input target**.

Therefore:

the black curve is the reconstruction target for the Autoencoder,

but it is **not an independent physical ground truth Nyquist curve**.

This distinction is very important.

The logic should be:

Measured EIS points → interpolation → AE input → AE reconstruction

Therefore, if the interpolation step is incorrect, good AE reconstruction would only mean that the AE successfully reconstructs an incorrectly processed signal.

This is why Steps 1–4 must be verified before making strong conclusions about the AE.

Step 6. Only after Steps 1–5 are confirmed, reevaluate the interpolation methods

Your current result shows that Log-Cubic actually gives the lowest test RMSE, while you selected Log-PCHIP because Cubic Spline produces visible overshoot.

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The rejection of Cubic Spline may be reasonable if the overshoot is confirmed after the raw data and frequency order are verified.

However, please do not write:

“PCHIP preserves the true physical shape.”

Instead, use wording such as:

“PCHIP was selected because it avoided the overshoot observed with Cubic Spline and better preserved the local shape of the measured EIS points.”

If possible, add a quantitative measure of interpolation distortion or overshoot rather than relying only on visual inspection.

Step 7. Correct the LOBO evaluation protocol

Currently, B0005 and B0006 are used for training, B0007 for validation, and B0018 as the unseen test battery.

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This is a valid held-out battery experiment, but it is not yet a complete Leave-One-Battery-Out evaluation.

A complete LOBO evaluation should let:

- B0005
- B0006
- B0007
- B0018

each serve as the unseen test battery in turn.

More importantly, B0018 has already been used repeatedly to compare interpolation methods, bottleneck sizes, L2 regularization, loss functions, and other choices. Therefore, B0018 is gradually becoming development data instead of a completely untouched test battery.

Please use:

Outer LOBO → final unseen-battery evaluation

and

Inner validation → model and hyperparameter selection

All design choices must be determined without using the outer test battery.

Step 8. Recheck the Causal Backward Mapping

Your tolerance = 3 selection is currently based mainly on the data retention rate:

- Tolerance 1: 50.96%
- Tolerance 2: 53.78%
- Tolerance 3: 99.89%

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This shows that tolerance = 3 retains almost all records, but it does not prove that tolerance = 3 gives the most accurate SOH labels.

Please compare:

Tolerance = 1, 2, and 3

using:

- Retention rate
- MAE
- RMSE
- R^2

Please also report the actual EIS-to-capacity cycle-gap distribution.

A three-cycle backward match does not cause future-data leakage, but it may still introduce label misalignment.

Step 9. Do not lock the AE bottleneck at 17

The PCA analysis shows that approximately 17 components retain 95% of the training-data variance.

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This is useful as an **initial candidate**, but it does not prove that the AE bottleneck should be 17.

Please revise:

“The bottleneck is locked at 17.”

to:

“PCA suggests 17 latent dimensions as an initial candidate based on 95% cumulative explained variance. The final AE bottleneck size will be selected according to inner-validation performance.”

Your own ablation study already shows that larger bottlenecks reduce training error but do not necessarily improve unseen-battery performance.

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Therefore, the final bottleneck must be selected by validation, not by PCA variance alone.

Step 10. Improve the AE reconstruction evaluation

The reconstruction RMSE values for train, validation, and B0018 are numerically quite close, while the visual Nyquist shapes appear very different.

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This means that a single absolute RMSE may not adequately represent EIS reconstruction quality.

Please add:

- $\text{Re}(Z)$ RMSE
- $\text{Im}(Z)$ RMSE
- Normalized RMSE
- Relative reconstruction error
- Mean $\pm$ standard deviation

for each battery.

After confirming the frequency ordering, please also show representative reconstruction plots from more than one cycle and more than one battery.

Step 11. Do not yet claim that “Correlation Flip proves the AE is the root cause”

The latent-feature correlation changes are interesting. For example, some latent nodes change significantly between the training batteries and B0018.

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However, this does not yet prove that the AE is the root cause.

Please change the conclusion from:

“The AE is proven to be the root cause.”

to:

“The AE latent representation exhibits instability under the unseen battery domain, which may contribute to the poor downstream prediction performance.”

For now, please call this:

latent feature instability

or

cross-battery domain shift

until further experiments establish causality.

Step 12. Figure 10 must be treated as a failed SOH/Capacity prediction result

Figure 10 is particularly important.

The ground-truth capacity clearly decreases with cycling, but the predicted result remains almost flat around a nearly constant value.

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The reported result is approximately:

- RMSE ≈ 0.1744 Ah
- $R^2 \approx -0.029$

This is not a successful degradation prediction.

The negative R^2 indicates that the model does not explain the capacity degradation trend better than a simple constant-mean prediction.

Please investigate why the prediction collapses toward a nearly constant value.

Do not describe bottleneck = 1 as the “best model” only because it gives the lowest RMSE.

A model with lower RMSE can still be useless for SOH estimation if it does not follow the degradation trajectory.

Step 13. Explain why Figure 10 collapses toward the mean

Please directly compare the Figure 10 prediction with a **Mean Predictor baseline**:

$$\hat{SOH} = \text{mean SOH of the training set}$$

or, if you are still using capacity:

$$\hat{Q} = \text{mean capacity of the training set}$$

Then compare:

- MAE
- RMSE
- R^2
- prediction trajectory

between:

Mean Predictor

and

AE-BPNN

If AE-BPNN performs similarly to the Mean Predictor, then the model has not learned meaningful degradation information.

For the paper, please avoid the informal term “Mean Prediction Trap.”

Use:

prediction collapse toward a near-constant value

or

regression toward the mean

instead.

Step 14. Compare Raw EIS, PCA, and AE before introducing Transfer Learning

I agree with your proposed three pipelines:

Raw EIS → BPNN

PCA → BPNN

AE → BPNN

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This experiment is now essential.

Please use exactly the same:

- Outer LOBO split
- Inner validation protocol
- Data mapping
- Input preprocessing
- SOH definition
- BPNN evaluation procedure
- Metrics

for all three pipelines.

Also add:

Mean Predictor

and preferably:

Linear Regression / Ridge Regression

as simple baselines.

This will answer an important question:

Does the AE actually improve SOH prediction, or does it make cross-battery generalization worse?

Step 15. Report SOH, not only Capacity in Ah

The research title is SOH estimation, but the current major performance results are mainly presented in Capacity (Ah).

Please clearly define:

$$SOH_k = \frac{Q_k}{Q_{\text{reference}}} \times 100\%$$

and explain how $Q_{\text{reference}}$ is determined for each battery.

The main results should then report:

- **MAE (%SOH)**
- **RMSE (%SOH)**
- **R²**

Capacity results in Ah can be reported additionally, but the main evaluation should be consistent with the research objective.

Step 16. Use Early Stopping instead of claiming that 300 epochs proves no overfitting

The current training curves indicate convergence at approximately 300 epochs. However, convergence does not prove the absence of overfitting.

Please use:

Early Stopping based on inner-validation loss

and save the model from the best validation epoch.

Different model configurations may require different numbers of epochs.

Step 17. Revise the Moving Average explanation

Please do not state that all capacity fluctuations are definitely caused by battery recovery during rest periods unless the NASA experimental records directly support this conclusion.

Use:

“The observed capacity fluctuations may be associated with recovery effects, measurement variability, or other experimental factors.”

At this stage, it is acceptable to retain the original unsmoothed labels.

The stronger justification is that we want to preserve the original measurements and maintain a causal real-time prediction setting.

Step 18. Required order for the next progress report

Please do not move to Transfer Learning yet.

For the next progress report, please confirm the issues in the following order:

- 1. Verify the original NASA EIS frequency information.**
- 2. Verify that the raw EIS points are ordered correctly by frequency.**
- 3. Redraw Figures 3 and 4 using the verified raw EIS information.**
- 4. Clearly identify measured data versus interpolated data and define the reference used in each figure.**
- 5. Reevaluate Linear, Cubic, and PCHIP only after the raw Nyquist data are confirmed.**
- 6. Recheck Figure 6 using the corrected preprocessing pipeline.**
- 7. Establish the complete nested four-fold LOBO protocol.**
- 8. Reevaluate the mapping tolerance using downstream MAE, RMSE, and R^2 .**
- 9. Compare Raw EIS $\rightarrow$ BPNN, PCA $\rightarrow$ BPNN, and AE $\rightarrow$ BPNN.**
- 10. Add Mean Predictor and Linear/Ridge Regression baselines.**
- 11. Report MAE (%SOH), RMSE (%SOH), and R^2 for every unseen battery.**
- 12. Explain Figure 10 quantitatively and determine why the prediction collapses toward a nearly constant value.**

13. Report the final results as mean $\pm$ standard deviation across the LOBO folds and, if possible, across multiple random seeds.

The most important point is:

Do not assume that the AE is the problem until the original EIS frequency information, Nyquist plotting, interpolation procedure, and evaluation protocol have all been verified.

The current poor result in Figure 10 may come from the AE, but it may also originate earlier in the pipeline. Therefore, we need to verify the entire chain step by step:

Raw EIS $\rightarrow$ Frequency Ordering $\rightarrow$ Interpolation $\rightarrow$ Feature Extraction $\rightarrow$ BPNN $\rightarrow$ SOH Prediction

Only after each stage is confirmed can we identify the true source of the generalization failure and decide whether Transfer Learning is really necessary.

資料來源